

ALFIDO TECH JAVA INTERNSHIP

TASK 3 : COLLECTIONS & EXCEPTION HANDLING

Objective

To understand Java Collections Framework and Exception Handling mechanisms.

Collections Used

ArrayList

Used to store ordered data.

HashMap

Used to store key-value pairs.

HashSet

Used to store unique values.

Exception Handling

Try Catch

Used to handle runtime exceptions.

Custom Exception

InvalidAgeException is created by extending Exception class.

Source Code

InvalidAgeException.java

```
package task3;

public class InvalidAgeException extends Exception {
    public InvalidAgeException(String message) {
        super(message);
    }
}
```

Main.java

```
package task3;

import java.util.ArrayList;
import java.util.HashMap;
import java.util.HashSet;

public class Main {

    public static void validateAge(int age) throws InvalidAgeException {
        if(age < 18) {
            throw new InvalidAgeException("Age must be 18 or above");
        }
    }

    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("Java");
        list.add("Python");
        System.out.println("ArrayList:");
        System.out.println(list);
        HashSet<String> set = new HashSet<>();
        set.add("Java");
        set.add("Java");
        set.add("Python");
        System.out.println("HashSet:");
        System.out.println(set);
        HashMap<Integer,String> map = new HashMap<>();
        map.put(1,"Shifani");
        map.put(2,"Golda");
        System.out.println("HashMap:");
        System.out.println(map);
        try {
            int result = 10 / 0;
        }
    }
}
```

```

        System.out.println(result);
    }
    catch(Exception e) {
        System.out.println( "Exception Caught: "+ e.getMessage());
    }
    try {
        validateAge(15);
    }
    catch(InvalidAgeException e) {
        System.out.println("Custom Exception: "+ e.getMessage());
    }
}
}

```

Screenshots

InvalidAgeException.java

```

1 package task3;
2
3 public class InvalidAgeException extends Exception {
4
5     public InvalidAgeException(String message) {
6         super(message);
7     }
8 }

```

Main.java

```

package task3;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.HashSet;

public class Main {

    public static void validateAge(int age) throws InvalidAgeException {
        if(age < 18) {
            throw new InvalidAgeException("Age must be 18 or above");
        }
    }

    public static void main(String[] args) {

        ArrayList<String> list = new ArrayList<>();

        list.add("Java");
        list.add("Python");

        System.out.println("ArrayList:");
        System.out.println(list);

        HashSet<String> set = new HashSet<>();

        set.add("Java");
        set.add("Java");
        set.add("Python");

        System.out.println("HashSet:");
        System.out.println(set);

        HashMap<Integer,String> map = new HashMap<>();

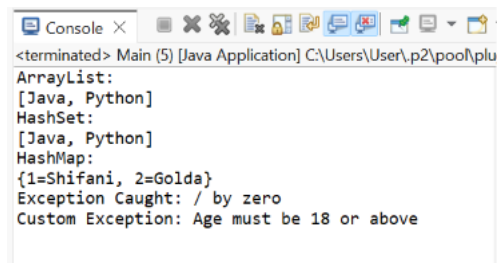
        map.put(1,"Shifani");
        map.put(2,"Golda");

        System.out.println("HashMap:");
        System.out.println(map);

        try {
            int result = 10 / 0;
            System.out.println(result);
        }
        catch(Exception e) {
            System.out.println( "Exception Caught: " + e.getMessage());
        }
        try {
            validateAge(15);
        }
        catch(InvalidAgeException e) {
            System.out.println("Custom Exception: " + e.getMessage());
        }
    }
}

```

Program Output



```

<terminated> Main (5) [Java Application] C:\Users\User\p2\pool\plu
ArrayList:
[Java, Python]
HashSet:
[Java, Python]
HashMap:
{1=Shifani, 2=Golda}
Exception Caught: / by zero
Custom Exception: Age must be 18 or above

```

Output

ArrayList:

[Java, Python]

HashSet:

[Java, Python]

HashMap:

{1=Shifani, 2=Golda}

Exception Caught: / by zero

Custom Exception: Age must be 18 or above

Conclusion

Successfully implemented ArrayList, HashMap, HashSet, try-catch blocks, and custom exceptions in Java.